

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm. Examination (Old Syllabus)

University Theory Examination Nov-Dec 2015

PH104 Pharmaceutical Engineering-I

Date: 21/12/2015, Monday Time:10:00 a.m to 01:00 p.m Maximum Marks: 80

Instructions:

- 1. Figures to right indicate marks.**
- 2. Section wise separate answer book to be used.**
- 3. Draw neat sketches wherever necessary.**

SECTION- I

Q 1 Attempt any TWO of the following.

- A Classify the types of flow. Describe the Reynolds experiment in detail. 10
- B Classify the flow meters. Give principle, working and application of Orificemeter with neat diagram. 10
- C Write a note on Solid handling systems in Pharmaceutical Industry. 10

Q 2 Attempt any TWO of the following .

- A Describe the various unit processes and unit operations. Explain the tie substance by giving suitable example. 10
- B Explain the various modes of mass transfer. Give brief note on solid-liquid mass transfer. 10
- C (1) Give various Colour coding of Pipelines. 05
- (2) Explain in brief dimensionless equation and formulas. 05

SECTION- II

Q 3 Attempt any TWO of the following.

- A Explain the various modes of heat transfer. Write a note on Fourier's law. 10
- B (1) Explain in brief the concept of Black body. 05
- (2) Explain steam as heating medium, properties and uses of steam, 05
- C Describe the various factors affecting the selection of material of pharmaceutical plant construction. 10

Q 4 Attempt any TWO of the following.

- A(1) Describe the various types of glass used in Pharmaceutical Industry. 05
- (2) Give the various types of corrosion with suitable example. 05
- B(1) Describe the various function and qualities of package. 05
- (2) Give type of packaging materials along with its merits and demerits. 05
- C Write a short note on Heat exchange equipments used in Pharmaceutical Industry. 10
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